

Thomas Schlaerth

Ace Vo

BSAN 6200

18 February 2026

Assignment 1 - Sentiment Analysis

Step 1:

The dataset being used today is a hotel reviews dataset taking 10,000 reviews of various hotels around the United States. What we are trying to accomplish is to run a sentiment analysis on the reviews to see whether they are positive or negative, and if they are negative, find the cause of the negative review. This analysis is done to highlight reoccurring issues for hotels and target actionable changes to improve overall guest experience.

Step 2:

Columns:

(['dateAdded', 'dateUpdated', 'address', 'categories', 'primaryCategories', 'city', 'country', 'latitude', 'longitude', 'name', 'postalCode', 'province', 'reviews.date', 'reviews.dateSeen', 'reviews.rating', 'reviews.text', 'reviews.title'])

Data Types:

The data types primarily consist of objects, and float64. Latitude, Longitude, and reviews.rating are float64 while all other columns are objects

Number of Rows:

The number of rows for this dataset is 10000

Missing Values:

The only columns that have missing values are reviews.text and reviews.title each with one missing value.

Main Text:

The column that contains the main text is reviews.text

Initial Observations:

Text is in English with no special characters such as emojis. Many reviews include numbers to indicate price, state abbreviations such as CA or MD, slang/text linguistics, and include contractions. Reviews also prominently feature an excessive use of punctuation and repetition of letters to emotion. There are some reviews with spelling errors which will need to be addressed. There are no HTML tags present in the reviews column. There are some obvious cleaning issues relating to standardizing text, removing punctuation, and removing white spaces.

Step 3:

Before and After Examples

Before: Our experience at Rancho Valencia was absolutely perfect from beginning to end!!!! We felt special and very happy during our stayed. I would come back in a heart beat!!!

After: our experience at rancho valencia was absolutely perfect from beginning to end! we felt special and very happy during our stayed. i would come back in a heart beat!

Before: VERY clean, comfortable place to stay located near downtown Napa. The grounds are nothing special, but there is a (very) small pool and jacuzzi, as well as a small fitness center. Clean, comfortable and conveniently located. If you plan to mostly be out and about in Napa, stay here. If you plan to spend your time lounging around the pool and grounds, pick someplace else. But you'll probably have to pay more. Thank you for your review and we are glad you enjoyed your stay.

After: very clean, comfortable place to stay located near downtown napa. the grounds are nothing special, but there is a very small pool and jacuzzi, as well as a small fitness center. clean, comfortable and conveniently located. if you plan to mostly be out and about in napa, stay here. if you plan to spend your time lounging around the pool and grounds, pick someplace else. but you will probably have to pay more. thank you for your review and we are glad you enjoyed your stay.

Before: Dirty rooms. Outdated decor. Faulty lights. Bad odor. Terrible breakfast. Would not recommend or stay here again. Disgusting.

After: dirty rooms. outdated decor. faulty lights. bad odor. terrible breakfast. would not recommend or stay here again. disgusting.

Step 4:

VADER Pipeline –

VADER handles punctuation, emojis, repeated characters

VADER has its own internal tokenizer which preserves repeated punctuation, looks out for capitalization, splits on whitespace and punctuation but keeps emojis, emoticons, slang, and internet shorthand

Minimal preprocessing can include removing HTML and trimming whitespace

Returns sentiment scores (neg, neu, pos, compound)

TextBlob Pipeline –

TextBlob is more flexible with standard cleaning procedures

Keep stop words and some punctuation

Returns polarity (-1 to +1) and subjectivity (0-1)

Transformer Pipeline –

DistilBERT handles text case, punctuation, emojis, emoticons, slang, and internet shorthand

When cleaning data remove HTML, whitespace, and officially tokenize with DistilBERT

Step 6:

--- VADER Success Examples ---

Example 1 (Original Index: 1):

Review Text: These little villas are surprisingly clean for the price. I liked that there was no carpet in our villa (Room 113- queen bed) because there's nothing worse than stepping on old, icky motel carpets. The floor was wood laminate that looked newly installed. The bathroom was fine except it did have a shower stall, so the space was a bit... More

VADER Prediction: positive

Ground Truth: positive

Justification: The VADER model successfully identified the sentiment.

Example 2 (Original Index: 2):

Review Text: A standard not fancy hotel that provides a little privacy and extra space at a good rate if you want an affordable suite. Decent pretty big breakfast buffet (could someone figure out that 15 of us don't digest dairy and really like having something like soy or almond milk). little pool only good for kids. Nice big lobby and lounge areas. Jesse L, Thank you for choosing to stay at Hyatt House during your recent trip to Seattle and for taking the time to share your experience on TripAdvisor. We value the feedback from our guests and are always striving to improve our guests stay. Please let us know when you return as we love to welcome you back as our guest. Sarah Junge

VADER Prediction: positive

Ground Truth: positive

Justification: The VADER model successfully identified the sentiment.

Example 3 (Original Index: 3):

Review Text: The Red Lion is owned by the same corporation that owns the Vineyard Inn, also in McMinnville. The Red Lion is a perfectly adequate hotel. The rooms are clean and the housekeeping adequate. The breakfast offered is OK--not gourmet, but gets you going in the morning. The Red Lion is a little quieter than the Vineyard Inn, since it is... More

VADER Prediction: positive

Ground Truth: positive

Justification: The VADER model successfully identified the sentiment.

--- TextBlob Success Examples ---

Example 1 (Original Index: 1):

Review Text: These little villas are surprisingly clean for the price. I liked that there was no carpet in our villa (Room 113- queen bed) because there's nothing worse than stepping on old,

icky motel carpets. The floor was wood laminate that looked newly installed. The bathroom was fine except it did have a shower stall, so the space was a bit... More

TextBlob Prediction: positive

Ground Truth: positive

Justification: The TextBlob model successfully identified the sentiment.

Example 2 (Original Index: 2):

Review Text: A standard not fancy hotel that provides a little privacy and extra space at a good rate if you want an affordable suite. Decent pretty big breakfast buffet (could someone figure out that 15 of us don't digest dairy and really like having something like soy or almond milk). little pool only good for kids. Nice big lobby and lounge areas. Jesse L, Thank you for choosing to stay at Hyatt House during your recent trip to Seattle and for taking the time to share your experience on TripAdvisor. We value the feedback from our guests and are always striving to improve our guests stay. Please let us know when you return as we'd love to welcome you back as our guest. Sarah Junge

TextBlob Prediction: positive

Ground Truth: positive

Justification: The TextBlob model successfully identified the sentiment.

Example 3 (Original Index: 3):

Review Text: The Red Lion is owned by the same corporation that owns the Vineyard Inn, also in McMinnville. The Red Lion is a perfectly adequate hotel. The rooms are clean and the housekeeping adequate. The breakfast offered is OK--not gourmet, but gets you going in the morning. The Red Lion is a little quieter than the Vineyard Inn, since it is... More

TextBlob Prediction: positive

Ground Truth: positive

Justification: The TextBlob model successfully identified the sentiment.

--- DistilBERT Success Examples ---

Example 1 (Original Index: 1):

Review Text: These little villas are surprisingly clean for the price. I liked that there was no carpet in our villa (Room 113- queen bed) because there's nothing worse than stepping on old, icky motel carpets. The floor was wood laminate that looked newly installed. The bathroom was fine except it did have a shower stall, so the space was a bit... More

DistilBERT Prediction: positive

Ground Truth: positive

Justification: The DistilBERT model successfully identified the sentiment.

Example 2 (Original Index: 2):

Review Text: A standard not fancy hotel that provides a little privacy and extra space at a good rate if you want an affordable suite. Decent pretty big breakfast buffet (could someone figure out that 15 of us don't digest dairy and really like having something like soy or almond milk). little pool only good for kids. Nice big lobby and lounge areas. Jesse L, Thank you for choosing to stay at Hyatt House during your recent trip to Seattle and for taking the time to share your experience on TripAdvisor. We value the feedback from our guests and are always striving to improve our guests stay. Please let us know when you return as we love to welcome you back as our guest. Sarah Junge

DistilBERT Prediction: positive

Ground Truth: positive

Justification: The DistilBERT model successfully identified the sentiment.

Example 3 (Original Index: 3):

Review Text: The Red Lion is owned by the same corporation that owns the Vineyard Inn, also in McMinnville. The Red Lion is a perfectly adequate hotel. The rooms are clean and the housekeeping adequate. The breakfast offered is OK--not gourmet, but gets you going in the morning. The Red Lion is a little quieter than the Vineyard Inn, since it is... More

DistilBERT Prediction: positive

Ground Truth: positive

Justification: The DistilBERT model successfully identified the sentiment.

--- Failure Examples ---

--- VADER Failure Examples ---

Example 1 (Original Index: 0):

Review Text: Everything was Ok But not great Staff seemed to not want to be there

VADER Prediction: negative

Ground Truth: neutral

Explanation of Failure: The VADER model's prediction of 'negative' did not align with the ground truth of 'neutral'. This could be due to nuances in language, sarcasm, or context not fully captured by the model.

Example 2 (Original Index: 13):

Review Text: Right off the interstate, over priced for the rooms and their current condition. Rug in the room stained and room smelled moldy. Checked out early....

VADER Prediction: neutral

Ground Truth: negative

Explanation of Failure: The VADER model's prediction of 'neutral' did not align with the ground truth of 'negative'. This could be due to nuances in language, sarcasm, or context not fully captured by the model.

Example 3 (Original Index: 23):

Review Text: We decided to book this hotel for a weekend stay-cation. We are from the Dayton area and knew the miller lane area had plenty of restaurants for our getaway. We have stayed in the south location by the Dayton mall and always enjoy our service, and the room... However, this location seems less organized and not as up kept. The employees parked wherever they wanted, maintenance trucks taking up half the parking lot (not even in spaces) and making it difficult since it was a one way driveway. Also, the bed was VERY hard and uncomfortable, I almost preferred the sofa pull out mattress. Our mirror was cracked, which wouldn't have been a big deal if we didn't have a toddler trying to touch it... And the tub was clogging when we showered. Also, the stove kept randomly turning on.. Yes seriously.. Just clicked on. Our sink

faucet had a steady drip as well as the bathroom cold knob did not turn (but the hot water also never got hot).

VADER Prediction: neutral

Ground Truth: negative

Explanation of Failure: The VADER model's prediction of 'neutral' did not align with the ground truth of 'negative'. This could be due to nuances in language, sarcasm, or context not fully captured by the model.

--- TextBlob Failure Examples ---

Example 1 (Original Index: 0):

Review Text: Everything was Ok But not great Staff seemed to not want to be there

TextBlob Prediction: positive

Ground Truth: neutral

Explanation of Failure: The TextBlob model's prediction of 'positive' did not align with the ground truth of 'neutral'. This could be due to nuances in language, sarcasm, or context not fully captured by the model.

Example 2 (Original Index: 13):

Review Text: Right off the interstate, over priced for the rooms and their current condition. Rug in the room stained and room smelled moldy. Checked out early....

TextBlob Prediction: positive

Ground Truth: negative

Explanation of Failure: The TextBlob model's prediction of 'positive' did not align with the ground truth of 'negative'. This could be due to nuances in language, sarcasm, or context not fully captured by the model.

Example 3 (Original Index: 28):

Review Text: This place has been renovated but not 100 percent. There are still some quirks and in some spots areas of improvement . The shower curtain is all plastic rubber and one of the

Windows lacked a curtain and only had slats that faced the street. My colleague said they couldn't sleep due to a loud fan going on and off... More

TextBlob Prediction: positive

Ground Truth: neutral

Explanation of Failure: The TextBlob model's prediction of 'positive' did not align with the ground truth of 'neutral'. This could be due to nuances in language, sarcasm, or context not fully captured by the model.

--- DistilBERT Failure Examples ---

Example 1 (Original Index: 0):

Review Text: Everything was Ok But not great Staff seemed to not want to be there

DistilBERT Prediction: positive

Ground Truth: neutral

Explanation of Failure: The DistilBERT model's prediction of 'positive' did not align with the ground truth of 'neutral'. This could be due to nuances in language, sarcasm, or context not fully captured by the model.

Example 2 (Original Index: 9):

Review Text: Bad: Delays in checkin, no liquor license yet!. Good: Location room

DistilBERT Prediction: positive

Ground Truth: negative

Explanation of Failure: The DistilBERT model's prediction of 'positive' did not align with the ground truth of 'negative'. This could be due to nuances in language, sarcasm, or context not fully captured by the model.

Example 3 (Original Index: 11):

Review Text: My daughter is a student at Virginia Tech, we had traveled down to attend farm day. The hotel is close, clean, good breakfast (except the eggs, they taste weird). Gotta be aware

even tho this is not a tourist area, there is a big difference between season and off-season rates, I did the same exact trip a month (june) ago and it was 40 less a night

DistilBERT Prediction: negative

Ground Truth: positive

Explanation of Failure: The DistilBERT model's prediction of 'negative' did not align with the ground truth of 'positive'. This could be due to nuances in language, sarcasm, or context not fully captured by the model.

The model I would recommend for this dataset is DistilBERT. Along with being the most accurate model, DistilBERT was able to understand the context of the sentiments and have the highest accuracy. Many reviews included a Bad and Good comment and DistilBERT was able to weight the options better than the other two models. The limitations with DistilBERT relate to the longer reviews and the bias from the model. Longer reviews can confuse the model due to it only being able to handle a certain number of characters. Bias from training also has an impact on the final analysis of the reviews. For longer text it would be better to choose VADER due to it having a pos, neg, and neutral score from a scale from 0-1. This would allow for calculations to be made according to the scores.